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Financial reinforcement learning under concept drift based on knowledge distillation and curriculum learning

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ABSTRACT

Market makers provide financial market liquidity by continuously offering buy and sell orders at publicly quoted prices, while simultaneously earning profits from the bid–ask spread in the process. Various deep reinforcement learning algorithms have been proposed to address such high-frequency sequential decision-making application. However, identifying and resolving the traditional concept drift problem of machine learning system in highly dynamic and complex financial environments has always been a very challenging task. In this paper, a novel reinforcement learning framework with environmental sentiment awareness incorporating curriculum learning and knowledge distillation is proposed. With the aid of a sudden concept drift detector based on market sentiment analysis, our trading model will restructure itself during significant market changes. Additionally, a novel curriculum learning method has been designed to enhance learning efficiency in diverse time segments comprising extensive learning environments. Furthermore, knowledge distillation is adopted to refine the agent's adaptive capabilities for handling daily gradual concept drift. Experiments with TAIEX Options (TXO) data demonstrate that our method outperforms traditional models, achieving a 38.17% increase in PnL-MAP and a 0.07 increase in Sharpe ratio, while maintaining comparable inventory risk. During testing, sudden concept drift events were detected approximately once every five market-making trading days (i.e., about once per week). This also validates that our proposed market-making strategy based on a sentiment-aware reinforcement learning framework effectively enhances trading performance by modeling sudden and gradual concept drifts.

1. Introduction

Reinforcement learning (RL) employs a trial-and-error approach to develop optimal trading strategies by interacting with financial markets [1]. This method is particularly crucial for high-frequency trading, such as market-making. Market makers enhance financial market liquidity by continuously quoting buy and sell orders at publicly quoted prices, while simultaneously earning profits from the bid–ask spread in the process. However, the rapidly changing market-making environment often leads to the problem of concept drift, which refers to unforeseen changes in the underlying distribution of market conditions over time. To tackle these challenges, we propose a novel reinforcement learning framework with environmental sentiment awareness, incorporating knowledge distillation and curriculum learning techniques, aimed at enhancing the trading model's adaptive capabilities for concept drift issues in the financial market.

In market-making, the application of deep reinforcement learning (DRL) has seen a notable increase, driven by its ability to uncover complex patterns in market data and deliver enhanced risk-adjusted returns compared to traditional analytical methods. Several studies have explored various reward functions to optimize the performance of market-making agents [2]. Additionally, there have been significant advancements in state representation and predictive modeling to enhance the decision-making capabilities of DRL agents [3,4]. Furthermore, enhancements to existing models through reinforcement learning methods have been proposed to create more adaptive and efficient market-making strategies [5].

However, in current high-frequency sequential decision-making applications based on DRL, identifying and resolving traditional concept drift problems of machine learning systems in highly dynamic and complex financial environments have not been adequately addressed. Concept drift can be classified into two main types: gradual concept drift and sudden concept drift. Gradual concept drift refers to slow, continuous changes in market behavior over time, while sudden concept drift represents sharp, abrupt changes triggered by major events

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or news. In order to adapt to daily gradual concept drift, enhancing the trading model's adaptive capabilities has become a crucial task. On the other hand, to mitigate losses caused by sudden concept drift, it is also very important to identify sudden changes and retrain the trading model efficiently.

To address the aforementioned issues, the initial development focused on creating a machine learning model for high-frequency market-making. This system must simultaneously address the practical and critical challenges of real-time system efficiency and effectiveness in complex financial markets, including sudden concept drift and gradual changes. For daily gradual changes, the knowledge distillation technique is adopted to intelligently guide the selection of historical environments for fine-tuning trading models. By refining the adaptive capabilities of the trading model, it becomes better equipped to handle new market situations. For sudden concept drift, the sudden drift detector and curriculum learning framework are further utilized to enhance learning efficiency across diverse time segments and extensive learning environments. This approach leverages market sentiment analysis to restructure the trading model promptly during significant market changes. Finally, to implement the details of the previously mentioned methods, time-series SSL was utilized to assist in the completion of market sentiment analysis, facilitating the effective realization of knowledge distillation and curriculum learning.

By effectively addressing concept drift, market makers are able to timely adjust their trading strategies, thereby enhancing trading efficiency. This plays a significant role in boosting the overall liquidity of the financial markets. Additionally, adapting to concept drift helps market participants handle potential high-risk situations with greater effectiveness. The key contributions of our work can be summarized as follows:

- A novel reinforcement learning framework with environmental sentiment awareness, incorporating knowledge distillation and curriculum learning, is proposed to address concept drift issues in financial trading strategies.
- Machine learning for high-frequency market-making must simultaneously address the practical and critical challenges of real-time system efficiency and effectiveness in complex financial markets, including sudden concept drift and gradual changes.
- The knowledge distillation technique is adopted to intelligently guide the selection of historical environments for fine-tuning trading models, thereby addressing daily gradual concept drift.
- The sudden drift detector and curriculum learning framework are further utilized to enhance learning efficiency across diverse time segments and extensive learning environments.
- The experimental results demonstrate that our method achieves better profitability, increasing PnL by 38.17% in TAIEX, while maintaining the same level of inventory risk.

2. Related work

The structure of this section is as follows: Section 2.1 discusses the evolution of market sentiment in finance. Section 2.2 examines the benefits of reinforcement learning (RL) and its application to financial trading. Finally, Section 2.3 analyzes the importance of knowledge distillation and curriculum learning.

2.1. Market sentiment in finance

Market sentiment reflects the overall attitude of investors towards a market, influenced by economic indicators, news, and psychological factors. It shows how investor emotions can significantly affect market prices, often causing overreactions or underreactions to events [6]. Market sentiment using natural language processing has gained significant attention, driven by its potential to provide real-time insights into investor behavior, predict market movements, and enhance trading strategies. Paramanik et al. [7] used text-based sentiment analysis to classify market sentiment and improve predictions of volatility. Shao et al. [8] proposed a dynamic multi-asset forecasting model that fuses LLM-based sentiment analysis with cross-sectional time-varying regression.

On the other hand, financial indicators have also been used as a basis for market sentiment. Joshie et al. [9] develop a dynamic standalone option selling strategy using the Put-Call Ratios and machine learning optimization. Sadorsky et al. [10] use random forests to predict the stock price direction of clean energy exchange-traded funds, utilizing Advance-Decline Lines as features. While previous studies have effectively simulated market sentiment using various financial indicators and machine learning methods, there remains a significant gap in their applicability to high-frequency trading. The real-time modeling of market sentiment, crucial for HFT environments, is often underexplored. These gaps are addressed by leveraging the buy-side put/call ratio, sell-side put/call ratio, and orderbook imbalance to simulate market sentiment within market-making. This ratio directly measures the balance between bullish and bearish outlooks.

2.2. DRL in financial trading

DRL's self-learning ability to derive optimal strategies from extensive data sets is particularly advantageous in financial trading. It leverages deep neural networks to process the high-dimensional, nonlinear data characteristic of financial markets, as explained by Sun et al. [11]. DRL strategies are notable for their end-to-end training, converting input features directly into trading decisions, while embedding trading objectives within the reward function to ensure alignment with specific trading goals. In the financial trading sector, DRL has been applied innovatively across various tasks, showcasing its versatility and efficacy. Eilers et al. [12] developed a self-learning trading system using RL and neural networks to optimize seasonal trading strategies. Kwon et al. [13] combined rule-based systems with DRL to create an adaptive trading strategy with dynamic volume control. Liu et al. [14] combined LSTM, DQN, and BERT for market sentiment prediction and trend analysis in the Chinese stock market. In the portfolio management domain, Vo et al. [15] integrated BiLSTM, ESG-based optimization, and RL to build socially responsible investment portfolios. Further enhancing DRL's capabilities in this area, Sang et al. [16] integrated multi-task self-supervised learning and adaptive exploration modules to enhance the generalization ability of the RL agent. Huang et al. [17] developed a competitive reinforcement learning system with fuzzy representation for strategy allocation. Exploring the realm of trading signals, Deng et al. [18] proposed a deep direct reinforcement model with a fuzzy representation for futures asset trading, and Liu et al. [19] applied imitative deep reinforcement learning to frequent minute trading.

Beyond financial applications, several DRL-based studies have addressed decision-making under complex or imbalanced conditions. Leng et al. [20] proposed a loosely coupled DRL framework for order acceptance, while Leng et al. [21,22] extended DRL with graph convolutional and dual-agent structures to enhance adaptability in dynamic environments. Similarly, recent studies have applied granular computing [23] and ensemble learning methods such as Random Forest [24] to improve performance on imbalanced datasets, providing complementary perspectives that could be integrated with financial reinforcement learning frameworks.

In terms of strategy improvement and market simulation, Kuo et al. [25] introduced LOB-GAN, an interactive environment designed to simulate trading interactions and enhance the generalization ability of trading agents. While previous studies have focused on enhancing model robustness through various innovative approaches, they often overlook the impact of rapid market changes and sentiment shifts on trading strategies. Our proposed framework not only includes a sudden drift detector to identify significant market changes, but also leverages the collaboration of knowledge distillation and curriculum learning to refine the trading model's adaptive capabilities for handling both gradual and sudden concept drift.

2.3. Knowledge distillation and curriculum learning

Knowledge Distillation(KD) [26] and Curriculum Learning(CL) [27] are two important techniques in machine learning. The core of knowledge distillation is the teacher–student framework. This paradigm involves a "teacher", which is a large, well-trained model, and a "student", a smaller and less complex model. The goal is to transfer knowledge from the teacher to the student, allowing the student to achieve comparable accuracy while significantly reducing computational complexity. Xu et al. [28] introduced the teacher-student Collaborative Knowledge Distillation method, which merges knowledge distillation with self-distillation for enhanced model performance, using a combination of learning from a teacher network and self-teaching within a student network, and applying an ensemble approach during inference. Chen et al. [29] employ a role-aware multi-agent system and a student–teacher framework to model volatile security markets and develop profitable trading strategies. The concept of CL was first introduced by Bengio et al. [30], drawing parallels between how humans and animals learn and how machine learning models could be trained. The core idea is that starting with easier examples and gradually increasing complexity can lead to more effective learning. This approach contrasts with traditional random exposure to training data [31], offering a more structured and efficient learning trajectory. Theoretical models suggest that this progressive approach can lead to faster convergence and better generalization [32]. Manela et al. [33] introduce an innovative algorithm that combines curriculum learning with Hindsight Experience Replay, specifically designed to facilitate learning in sequential object manipulation tasks within multi-goal and sparse feedback scenarios.

While previous studies have made significant strides in using curriculum learning and knowledge distillation to enhance model training, they each have certain limitations when dealing with concept drift. In our framework, we bridge these gaps by employing a hybrid method that utilizes both curriculum learning and knowledge distillation to enhance the trading model's adaptive capabilities for concept drift issues in the financial market.

3. Methodology

In this section, the system architecture and design details of market-making are outlined in . The implementation details of the Time-series Self-Supervised Market Sentiment Network (TSSMSN) are explained in Section 3.2. Section 3.3 introduces the training process of RL for market-making. Details of the knowledge distillation process are provided in Section 3.4. Finally, Section 3.5 offers detailed information about curriculum learning.

3.1. System overview

Motivated by [34], we proposed a reinforcement learning framework with environmental sentiment awareness for market-making, which is depicted in the system overview diagram in Fig. 1. The first section is the TSSMSN. The goal is to extract real-time market sentiment to effectively realize subsequent downstream tasks. The second section describes the reinforcement learning framework with environmental sentiment awareness, which serves as an efficient high-frequency trading decision system. The third section is the knowledge distillation and the curriculum learning. Once deployed, the trading model will constantly encounter market changes. Through retraining or fine-tuning promptly, our trading model will restructure itself during concept drift. In the following sections, the operation of each block will be detailed. Next, we will introduce the problem definition, outline the market-making procedure, and discuss the limitations of our framework.

3.1.1. Problem formulation

The process of placing orders in market-making can be conceptualized as a Markov Decision Process (MDP). This framework is particularly apt for market-making, as it involves a sequence of decisions – at what price, and in what quantity – made in an environment that is continuously changing and inherently uncertain. The state (S) represents specific market conditions, encapsulating factors such as the current orderbook status, recent trades, and price movements. The actions (A) involve strategic decisions like placing buy or sell orders, guided by various market-making strategies. The transition probability (P) reflects the stochastic nature of market dynamics, influenced by the actions of the market maker and external factors like price fluctuations and incoming orders. The complete process of RL can be represented as $\langle S, A, P, R\rangle$. The framework is as follows: first, the agent observes the market state $s_t \in S$ at time t, and based on this, takes an action $a_t \in A$, such as placing a buy or sell order. The market responds to this action, leading to a reward (or penalty) for the agent, computed through a reward function as $r_t = R(s_t, a_t, s_{t+1})$.

The market transitions to a new state s_{t+1} according to the transition probability $P(s_{t+1}|s_t, a_t)$. The agent's goal is to learn an optimal policy π, which dictates the best action to take in any given market state to maximize the expected total return. This involves balancing immediate rewards and future state benefits, effectively optimizing for both profit maximization and inventory risk minimization. The optimal policy π is defined as $\pi^* = \mathbb{E}_{p(\tau)}\left[\sum_{t=1}^{T} \gamma^{t-1} r_t\right]$, where $\gamma \in [0, 1)$. The discount factor, denoted as γ, is applied to the trajectory, which consists of a sequence of states and actions. When the discount factor is set to a smaller value, the agent tends to prioritize immediate rewards. Conversely, a larger discount factor encourages the agent to adopt a longer-term perspective, favoring decisions that yield greater long-term benefits. We provide a more detailed explanation of our choice of the discount factor in the experimental section. The trajectory $\tau = \left\{(s_1, a_1), (s_2, a_2), \ldots, (s_T, a_T)\right\}$ presents the formulation of this trajectory.

3.1.2. Procedure for market-making

The approach adopted involves a dynamic strategy for market-making, where order placements are based on changes in the Limit Order Book (LOB) rather than on a rigid time schedule. As illustrated in Fig. 2, our agent repositions its orders after every 20 LOB modifications. This strategy is tailored to the fluctuating nature of market activity, allowing for more responsive and effective trading. Contrast this with a fixed-interval method, like placing orders every 5 s, which can be less effective. In times of low market activity, such a rigid schedule could result in minimal trading, leading to sparse rewards and potential instability in the trading strategy.

The process begins with the agent determining its order placement relative to the current best bid–ask price. The order remains active until it is either executed or 20 changes in the LOB occur. Execution happens when the best bid or ask price meets the order price. The model is simplified by assuming the order is always last in the queue and restricting the order size to a single lot. Regardless of whether the order is executed, a reward is calculated. This marks the end of one cycle and sets the stage for placing the next order. This study's examination of market-making is based on certain assumptions and constraints, which include the following:

- Our study overlooks the probability of market orders and the lineup of limit orders, assuming the market maker's orders are always last in the orderbook. A trade happens when the market price matches the limit order price.
- We do not consider the number of executed trades; we assume that each trade is carried out with a standard size of one lot.
- Our model does not factor in transaction costs, meaning each trade is executed without incurring any extra fees or taxes.
- Our study also assumes negligible microsecond-level latency in high-frequency trading, including delays caused by network transmission, order matching systems, and algorithm execution.

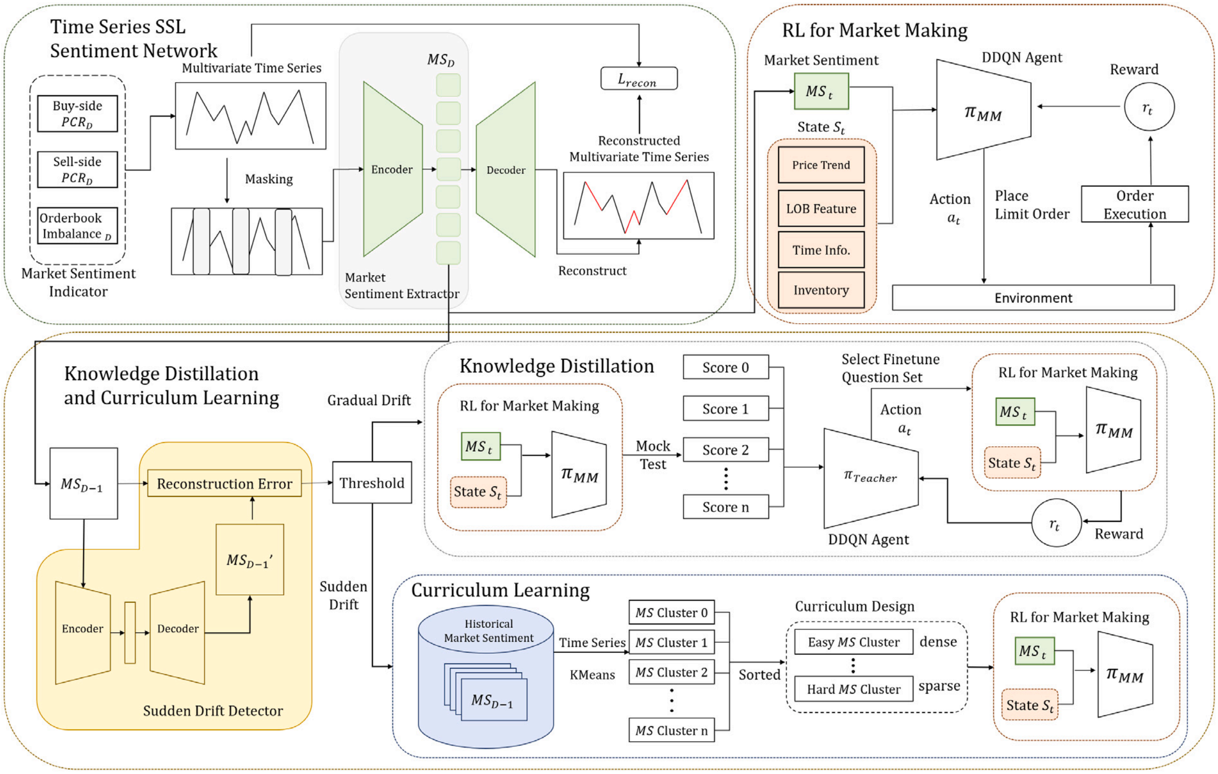

Fig. 1. System overview of sentiment-aware RL for market-making.

Fig. 2. Market making procedure.

Market makers of derivative financial products in Taiwan must obtain approval from the competent authority and are eligible for transaction and clearing fee incentives granted by the exchange. Those who fully meet their obligations and achieve specified performance benchmarks may receive up to a 100% reduction in trading costs. In addition to fee reductions, market makers may also benefit from revenue-sharing arrangements, relaxed trading restrictions, and exclusive product distribution rights, thereby gaining additional profitability and competitive advantages. As in previous literature [2], the market-making strategy can therefore be reasonably modeled under a zero transaction cost assumption. These assumptions draw from previous studies, but may not fully reflect real trading situations. Caution is advised when interpreting this study's results. Despite these constraints, our research provides valuable insights into using advanced deep learning methods for smarter market-making strategies.

3.2. Time-series self-supervised market sentiment network

In financial markets, investors' behavior and decisions are often influenced by emotions, which is collectively known as market sentiment. Market sentiment can significantly impact market trends. In many current studies, incorporating market sentiment into strategy design has proven to be an effective approach. Therefore, integrating market sentiment into models can provide a more comprehensive market perspective, helping the model better understand market trends and make more informed decisions.

However, effectively quantifying market sentiment is a critical and challenging issue. We drew inspiration from CNN's Fear and Greed Index, which references several different market indicators, such as volatility, stock price strength, stock price momentum, and market participation. These indicators collectively depict the emotional state of market participants. The index helps us understand the direction of market sentiment and further assists in decision-making. Based on this concept, the ideas of the put/call ratio and orderbook imbalance were incorporated into the model, making them the core components of the market sentiment indicator. This combination is believed to provide a more accurate and in-depth assessment of market sentiment in market-making environments, thereby improving the model's performance.

3.2.1. Market sentiment indicator

The core of our market sentiment indicators centers on the put/call ratio (PCR) and orderbook imbalance (OI), both designed to assess the market's directional bias through options trading. This section elaborates on the computation methods for two distinct PCR variants: $\mathrm{PCR_B}$, reflecting buyer sentiment, and $\mathrm{PCR_S}$, reflecting seller sentiment. Alongside the PCR metrics, we introduce and elaborate on the computation methodology and the significance of the OI.

- **Buy-side Put/Call Ratio** ($\mathrm{PCR_B}$): The buy-side put/call ratio is computed by dividing the quantity of put options purchased by the quantity of call options purchased. The formula is given as follows: $PCR_{\mathrm{B}} = \frac{\mathrm{PUT}_{Q_{\mathrm{Buy}}}}{\mathrm{CALL}_{Q_{\mathrm{Buy}}}}$, where $\mathrm{PUT}_{Q_{\mathrm{Buy}}}$ represents the volume of put options bought and $\mathrm{CALL}_{Q_{\mathrm{Buy}}}$ denotes the volume of call options bought. A higher $\mathrm{PCR_B}$ suggests a bearish sentiment among buyers, indicating a predilection for purchasing puts over calls, often in anticipation of a market downturn. Conversely, a lower $\mathrm{PCR_B}$ indicates a bullish sentiment, suggesting that buyers are more inclined towards acquiring call options, expecting the market to rise.

- **Sell-side Put/Call Ratio** ($\mathrm{PCR_S}$): The sell-side put/call ratio is determined by the ratio of the quantity of put options sold to the quantity of call options sold. The calculation is expressed as $PCR_S = \frac{\mathrm{PUT}_{Q_{\mathrm{Sell}}}}{\mathrm{CALL}_{Q_{\mathrm{Sell}}}}$. In this context, $\mathrm{PUT}_{Q_{\mathrm{Sell}}}$ is the volume of put options sold, and $\mathrm{CALL}_{Q_{\mathrm{Sell}}}$ is the volume of call options sold. A higher $\mathrm{PCR_S}$ reflects a bullish sentiment among sellers, showing a tendency to sell puts over calls, which may suggest an expectation of a market rise. On the flip side, a lower $\mathrm{PCR_S}$ points towards a bearish sentiment among sellers, with a greater disposition towards selling calls over puts, anticipating a market decline.
- **Orderbook Imbalance (OI)**: The formula calculates short-term buying and selling pressure by subtracting the sum of the top five bid volumes from the sum of the top five ask volumes and then dividing by the total volume. This is expressed as $OI = \frac{\sum_{i=1}^{5}(V_i^A - V_i^B)}{\sum_{i=1}^{5}(V_i^A + V_i^B)}$, where V_i^A and V_i^B represent the ask and bid volumes at different price levels. A high OI value indicates a predominant buying pressure in the market. This situation often reflects a bullish sentiment among traders, suggesting that there are more buyers than sellers at the top of the orderbook. Traders might interpret this as a potential upward movement in the asset's price. Conversely, a low OI value signifies a dominant selling pressure. This condition is indicative of a bearish sentiment, where the number of sellers outweighs the number of buyers at the top levels of the orderbook. Market participants might view this as a sign that the asset's price could experience a downward trajectory.

Then, we introduce a method to capture market sentiment at any given moment through a feature set composed of multiple distinct values. This set incorporates $\mathrm{PCR_B}$, $\mathrm{PCR_S}$, and OI, each assessed over a look back window size of market sentiment (LB_{MS}) period. The formula for each moment's feature set can be represented as Eq. (1).

$$MS_t = \left\{ PCR_{B,t-i}, PCR_{S,t-i}, OI_{t-i} \mid i = 0, 1, \dots, LB_{MS} \right\} \tag{1}$$

In Eq. (1), the market sentiment at any given time t, denoted as MS_t, is formulated by incorporating a set of three pivotal metrics, each evaluated over a sequence of ten preceding time slots. Each of these indicators is appraised at time intervals $t-i$, where i ranges from 0 to LB_{MS}.

3.2.2. Time series SSL with autoencoder

In terms of time series self-supervised methods, our choice leans towards generative-based methods, with a special focus on employing autoencoder-based reconstruction. Autoencoders serve as a powerful tool in unsupervised learning, designed to encode input data x into a compressed latent space representation z and subsequently reconstruct the input from this representation, the reconstructed input is defined as $\hat{x}$. The goal is to minimize the difference between x and $\hat{x}$, typically quantified by a loss function $\mathcal{L}(x, \hat{x}) = \frac{1}{N}\sum_{i=1}^{N} \left\| x_i - \hat{x}_i \right\|^2$.

The use of a mask autoencoder for feature extraction in multivariate time series was explored. The principle of a mask autoencoder is rooted in selectively masking portions of the input data before passing them through the encoder. This approach forces the encoder to learn more robust and comprehensive representations of the data, as it has to predict the masked parts without direct access to the full input. First, a portion of the input time series data is randomly masked, creating a partial view of the data. The masked data is then encoded into a latent representation z and subsequently decoded to reconstruct the original unmasked data $\hat{x}$. The objective remains to minimize the reconstruction loss between the original data x and the reconstructed data $\hat{x}$, employing a loss function $\mathcal{L}_{recon} = \sum_{i=1}^{N} Mask_i \cdot (x_i - \hat{x}_i)^2$. Where $Mask_i$ is the mask value at the ith data point; it is a binary indicator (0 or 1) that determines whether the corresponding data point contributes to the loss function.

Building on the framework established by the mask autoencoder for feature extraction in multivariate time series, the trained model's encoder has been leveraged for dimensionality reduction. Specifically, the encoder is employed to distill the essence of the data-rich in market sentiment derived from market-making, denoted as MS_t, into a more compact and informative feature space. By processing MS_t through the encoder, the most salient features are captured into a reduced set of dimensions, effectively encapsulating the critical information content of the original data while significantly reducing its complexity.

3.3. Reinforcement learning in market-making

In this section, we will explain the state, action, and reward components utilized in RL for market-making. By defining the state space, we can capture the essential information about the market environment. The action space delineates the possible decisions the market maker can take, while the reward function quantifies the profitability of these actions.

The state for high-frequency market-making tasks encompasses several key elements: orderbook data, options pricing indicators, product information for options, and inventory holdings. Details on each of these aspects are provided below.

- **Mid price return**: Indicates the variation in the mid-price in a high-frequency context, which is the average of the best bid and ask prices. This characteristic maintains a sequence for a past period, defined by the length $\mathrm{lookback}^{\mathrm{state}}$.
- **Minute-level price return**: To capture longer-term price trends, the opening, high, low, and closing prices from the last 5 min, all normalized by the current mid-price, are incorporated. This feature maintains a historical sequence with a length of $lookback^{state}$.
- **Implied volatility**: A key factor in option pricing, is derived using the bisection method with the Black–Scholes formula. This method iteratively narrows down the volatility range by comparing the formula's estimated price with the market price.
- **Log moneyness**: When an option becomes significantly in-the-money or out-of-the-money, trading interest often decreases.
- **Inventory**: The agent's order strategy is significantly influenced by its current inventory, measured in lots and subject to an inventory limit. This setup allows for negative inventory positions, indicating short contracts. The inventory is standardized by dividing the current position by the limit, keeping it within a $[-1, 1]$ range.
- **Time to market close**: Market activity often peaks during opening and just before closing. Recognizing the remaining time until market close can aid an agent in refining its strategy, like adjusting spreads and identifying cyclical patterns.
- **Time to maturity**: Options nearing expiration typically hold lesser time value, impacting market pricing and trading behavior. For our study focusing on options expiring within a month, we normalize time to expiration with 30 days as the base measure.

In a finite discrete action space $A = \{a_0, a_1, a_2, \dots, a_7\}$, each action $a = (\zeta_A, \zeta_B)$ denotes a pair of ask–bid spread. Where ζ_A is the number of ticks from the best ask price, and ζ_B is the number of ticks from the best bid price. Upon executing an order, these values are scaled by the tick size ζ_{tick} to determine the buying and selling prices, as detailed in Eq. (2).

$$\begin{aligned} P^{Ask} &= P_0^A + \zeta_A * \Delta_{tick} \\ P^{Bid} &= P_0^B + \zeta_B * \Delta_{tick} \end{aligned} \tag{2}$$

Table 1
Action space for market making.

Action ID	a_0	a_1	a_2	a_3	a_4	a_5	a_6	a_7
Ask ζ_A	0	0	+1	+1	+2	+2	+3	+3
Bid ζ_B	0	−2	−1	−3	0	−2	−1	−3

The tick size in the options products we use, represented by Δ_{tick}, is adjusted based on the options' price. This variation helps to avoid creating an excessively large action space, which could otherwise impede the efficiency of the model's convergence. Specifically, for options priced below \$10, the tick size is set at \$0.1. As the price increases, the tick size adjusts accordingly: it is \$0.5 for prices ranging from \$10 to just under \$50, \$1 for those from \$50 to below \$500, \$5 for prices within the \$500 to under \$1000 bracket, and for options priced at \$1000 or above, the tick size is \$10. This tiered structuring of tick sizes ensures a balanced action space across different price ranges. The range of action is detailed in Table 1. Consider an instance of placing an order: suppose the current highest bid price is \$450 and the lowest ask price is \$452, with a tick size of \$1. If the selected action is $a_3 = (+1, -3)$, the market maker would set their quote at \$447 for buying and \$453 for selling. The spread skew is employed to manage inventory risk. For instance, if the market maker currently holds a significant long position, they are more inclined to quote using a_1 or a_3. This strategy increases the likelihood of selling assets rather than buying, helping decrease the inventory level. In market-making, the main goal is to maximize strategy profitability while keeping inventory accumulation in check. There are two primary ways to make a profit. The first is through capturing the bid–ask spread. Specifically, when the market maker's sell order is executed above the mid-price or the buy order below it, the corresponding profits can be expressed as ϕ_t^a and ϕ_t^b in Eqs. (3) and (4). Their sum, as shown in Eq. (5) represents the total spread profit, which ideally occurs without changing the inventory level. The second profit source comes from the appreciation in the value of holdings due to market price movements. While holding larger inventory can lead to significant gains, it is a risky approach, as swift price fluctuations can result in losses. A market-neutral and conservative market maker would prefer to keep a minimal inventory, focusing on earning from the spread via continuous, two-sided trading. Our reward equation, as shown in Eq. (6), includes ϕ_t for bid–ask spread profits, q_t for inventory level, p_t^{mid} for the current mid-price, and Δm_t for the change in mid-price. Regarding the reward design in our RL framework, there are two main ideas behind it. For a market maker, maintaining a large one-sided inventory is generally undesirable, as it can lead to catastrophic losses when prices fluctuate. Therefore, we incorporate inventory risk $\epsilon\|q_t\|$ into the reward function. This penalty operates such that as the agent's inventory position grows, the negative reward it incurs increases as well. Consequently, the agent is incentivized to actively reduce its inventory. The goal is to encourage profit generation through well-balanced bid–ask placements (i.e., market-making profits) rather than accumulating excessive one-sided positions that increase unrealized P&L. Moreover, the penalty cost term follows the setting in prior work [2], which conducted a sensitivity analysis to evaluate how this term influences the overall performance. It is worth noting that, for the sake of simplicity, we assume that transaction costs are negligible, given that market makers typically receive rebates from the exchange.

$$\phi_t^a = \begin{cases} p^{ask} - p_t^{mid} & \text{if matched} \\ 0 & \text{otherwise} \end{cases} \tag{3}$$

$$\phi_t^b = \begin{cases} p_t^{mid} - p^{bid} & \text{if matched} \\ 0 & \text{otherwise} \end{cases} \tag{4}$$

$$\phi_t = \phi_t^a + \phi_t^b \tag{5}$$

$$R(t) = \frac{\phi_t + q_t \cdot \Delta m_t}{\Delta_{\text{tick}}} - \epsilon \cdot \|q_t\| \tag{6}$$

3.4. Gradual concept drift and knowledge distillation

Knowledge distillation is adopted to refine the trading model's adaptive capabilities for handling daily gradual concept drift. It involves a teacher model guiding a student model's learning process through strategic training decisions, thereby addressing daily gradual concept drift. In this section, the exploration focuses on how the teacher model guides the trading model's learning using reinforcement learning.

At any decision point, the teacher model assesses the state S_t, which is based on evaluating the market-making trading model's performance indicators from a mock test. These indicators include PnL and MAP. The mock test consists of ten questions, selected to represent the centroid of each cluster in a set. These centroids act as representative questions for their respective clusters. Thus, the state representation S_t integrates these performance metrics across the ten selected centroid questions. Consequently, at each time point, the state comprises twenty elements (10 mock test questions multiplied by 2 performance indicators: PnL and MAP), as shown in the Eq. (7).

$$S_t = (PnL_i^{mock}, MAP_i^{mock})_{i=1}^{10} \tag{7}$$

The action A_t taken by the teacher model at any decision point involves selecting one of the 10 question sets for the market-making model to train on. Thus, the action space is discrete, with 10 possible actions corresponding to the question set choices. The Action space A_t is shown as Eq. (8).

$$A_t = \{a_t | a_t \in \{QS_1, QS_2, \dots, QS_{10}\}\} \tag{8}$$

In this equation, a_t represents the specific action taken at time t, which entails choosing one question set from the set $QS_1, QS_2, \dots, QS_{10}$. The selection of the question set is critical as it directly influences the performance of the market-making model. The teacher model must consider various factors, such as the diversity of the topics and the current performance level of the market-making model. By strategically selecting the most appropriate question set at each decision point, the teacher model aims to optimize the training efficiency and overall performance of the market-making model.

The reward structure of the teacher model is intricately designed to enhance the learning capabilities of the market-making trading model by focusing on changes in PnL. This is done by comparing the PnL values before and after training through a series of ten mock tests calculating the change. The average improvement in PnL, denoted as Avg(ΔPnL), serves as a primary component of the reward function, reflecting the market-making model's performance improvement.

In addition to performance-based rewards, the reward function incorporates a penalty to discourage the repetitive selection of training datasets. If the teacher model chooses a dataset previously selected in other iterations, a penalty is applied. This penalty is calculated based on the frequency of dataset selection, using a coefficient λ and an exponent α, which together determine the intensity of the penalty for repeated selections. The purpose of incorporating this penalty is to encourage the exploration of a diverse range of training materials. This helps prevent the teacher model from over-relying on specific datasets, thus avoiding overfitting and promoting a broader learning experience for the market-making model. The complete reward function, labeled as R_{t+1}, is given by Eq. (9):

$$R_{t+1} = \text{Avg}(\Delta\text{PnL}) - \lambda \times (Num_{Repeat})^{\alpha} \tag{9}$$

In the reward function R_{t+1}, specified for the teacher model at time step $t+1$, the term Avg(ΔPnL) measures the average improvement in PnL. The penalty term, $\lambda \times (Num_{Repeat})^{\alpha}$, imposes a cost for repeatedly selecting the same dataset. The coefficients λ and α are crucial as they regulate the penalty's severity and its growth rate with increasing repetitions, respectively. The penalty design is inspired by entropy regularization, aiming to avoid over-concentration and encourage diverse strategies. Initially, we experimented with penalizing repeated curriculum selections in the teacher model. Empirical results showed that an exponential-shaped penalty was more effective than a linear form in suppressing overly concentrated behaviors. Therefore, we adopted this version to balance exploration and exploitation, ensuring more diverse curriculum usage during fine-tuning. This reward structure is designed to strike a balance between reinforcing beneficial learning patterns and promoting a diversity of training inputs, ultimately enhancing the learning efficacy of the market-making model.

3.5. Sudden concept drift and curriculum learning

Curriculum learning has been designed to enhance learning efficiency in diverse time segments comprising extensive learning environments. In this section, we present two key components: a sudden drift detector to identify abrupt changes in market sentiment and a time series k-means clustering strategy that underpins a tailored curriculum learning approach.

3.5.1. Sudden drift detector

To address the issue of identifying significant market changes in dynamic markets, a methodology employing an autoencoder-based detector for sudden concept drift in MS is proposed. The essence of this approach lies in training an autoencoder on historical MS data to capture the underlying patterns and regularities of market sentiment.

The methodology is structured as follows: Initially, the autoencoder is trained on a dataset comprising historical MS values, enabling it to learn a representation of normal market sentiment fluctuations. The training process involves minimizing the reconstruction error, typically measured by the mean squared error (MSE) between the original data X and the reconstructed data $\hat{X}$. Upon encountering new market conditions, the autoencoder evaluates the reconstruction error to detect sudden concept drifts. A significantly higher error than the threshold established during training suggests a deviation from the learned patterns, indicating significant market changes. This implies that the market sentiment has undergone a substantial change, necessitating a retraining of the market-making model to adapt to the new market dynamics.

3.5.2. Time series k-means clustering and curriculum strategy

We introduce an approach to training RL for market-making, employing a curriculum learning strategy informed by market sentiment MS. It consists of clustering of market sentiment data into groups, and a curriculum learning strategy that organizes these groups by their complexity for the RL model. To facilitate the curriculum learning approach, we apply time series k-means clustering to segment the market sentiment data into N_{cur} distinct groups, based on the similarity of sentiment patterns over time. N_{cur} represents the number of curriculum, which denotes the number of different learning stages used in the framework. The formulation for this clustering process is:

- Let $\mathbf{X} = \{x_1, x_2, \dots, x_n\}$ denote the set of multivariate time series data, where each x_i represents a time series of length 1000 with 10 dimensions.
- The objective of time series k-means clustering is to partition $\mathbf{X}$ into k clusters $\mathbf{C} = \{C_1, C_2, \dots, C_k\}$ in a way that minimizes the within-cluster sum of squares (WCSS).

The curriculum learning strategy is designed around the concept of progressively training the RL model on tasks of increasing complexity. This complexity is determined by the cohesion within each sentiment cluster:

- Clusters with tighter, more cohesive sentiment patterns are considered simpler learning tasks because they represent more consistent market conditions.
- Clusters with more dispersed sentiment patterns represent complex learning tasks due to their indication of volatile market conditions.

To quantify the cohesion of each cluster, we can use the within-cluster variance as a measure. This metric helps to identify how closely related the data points within a cluster are, reflecting the cluster's homogeneity. The lower the within-cluster variance, the more cohesive the cluster is. The cohesion of a cluster, indicating its tightness or the similarity of its elements, is quantified using the within-cluster variance. This variance measures the average squared distance from each point in the cluster to the cluster centroid, serving as a proxy for the cluster's homogeneity. This method combines time series k-means clustering with a curriculum learning strategy in an RL framework for market-making. By clustering sentiment data into ten groups based on temporal dynamics and organizing these groups by complexity, the RL model progressively learns from stable to volatile market conditions. Complexity is assessed by within-cluster variance, guiding the curriculum to train the trading model for diverse market scenarios effectively.

For example, in our experiments, clusters with lower within-cluster variance typically correspond to stable market conditions, such as when the PCR remains within a relatively fixed range and the OI shows little fluctuation. In contrast, clusters with higher within-cluster variance represent more volatile market conditions, where PCR fluctuates drastically and the OI frequently shifts between positive and negative values. Through curriculum learning, the agent first trains on stable clusters and then gradually progresses to volatile clusters, ultimately enabling the model to better adapt to diverse market conditions.

After introducing all of the modules, Alg. 1 presents the CL&KD pseudocode, demonstrating the integration of knowledge distillation and curriculum learning to dynamically adapt the market-making RL model. The pseudocode outlines the process of detecting sudden drifts in market sentiment, clustering market sentiment data for curriculum learning, and utilizing the teacher RL model for knowledge distillation. This dual approach ensures that the RL model can effectively respond to both gradual and abrupt changes in market conditions, enhancing its adaptability and performance in high-frequency trading environments.

Algorithm 1 Training Procedure for teacher RL Model

Require: SD: Sudden Drift Detector, $MMRL$: Market Making RL Model, TRL: Teacher RL Model, MS: Market Sentiment
Ensure: $MMRL\ \pi'$: Adapted Market Making RL Model
for days in testing period **do**
  Check for sudden drift in $MS[d-1]$ using SD
  if sudden drift is True **then**
    Divide MS into N clusters
    Compute the within-cluster variance for each cluster to measure its compactness
    Rank clusters by within-cluster variance
    Initialize the $MMRL\ \pi_0$
    for each cluster i from 1 to N **do**
      Retrain the $MMRL\ \pi$ on Cluster i
    end for
    Return: Retrained $MMRL\ \pi'$
  else
    for each guide iteration **do**
      $MMRL\ \pi$ performs a mock test consisting of ten questions
      TRL observes the performance(PnL,MAP) of π on the mock test
      TRL selects a question set QS for MMRL π
      $MMRL\ \pi$ fine-tunes on the selected question set QS
    end for
    Return: Fine-tuned $MMRL\ \pi'$
  end if
end for

4. Experiment

In this section, we begin by presenting the dataset, experiment setting, evaluation metrics, and comparative methods. To substantiate the effectiveness of our approach and to analyze the experimental results, four research questions are presented and addressed individually following Section 4.5. Here are our research questions:

- **RQ1:** How does varying the threshold for sudden drift detector and the frequency of model updates influence the model's overall performance?
- **RQ2:** Does our method outperform other control groups?
- **RQ3:** Which market sentiment information enhances RL performance in market-making?
- **RQ4:** Does our method outperform others when applied to put options?

4.1. Dataset and experimental settings

We utilized tick data from the TAIEX Options (TXO) on the Taiwan Futures Exchange (TAIFEX), focusing on the January 2017 to June 2019 period. Our analysis segmented the data into a training set (January 2017 to December 2018) and a test set (January to June 2019), excluding the expiration day's data to avoid volatility spikes, yielding 469 training and 109 testing days. Our study concentrated on tick-level limit orderbook data, capturing the top five bids and asks with volumes from market open to close. We excluded entries from the initial and final five minutes daily to mitigate anomalies, focusing on 8:50 am to 1:40 pm trading data. The dataset zeroes in on the two most traded TXO contracts daily, amassing 938 and 218 distinct training and testing environments, respectively, averaging over 31,000 orderbook records per day. We chose Elegantrl [35] as our RL framework due to its efficient training capabilities and extensive evaluation tools, which are ideal for scholarly inquiries. For our policy, we implemented the Double Deep Q-Network (DDQN) [36], a reliable and proficient critic-based approach for environments with discrete action choices. In addition, the discount factor controls the relative importance of long-term rewards versus short-term profits. Various settings were tested in our proposed trading system. After extensive experimentation, we found that since the overall trading horizon in our setup is relatively short, different discount factors led to only about a $\pm 2\%$ variation in rewards, indicating a limited impact on overall performance. The final value of the discount factor was empirically selected through a trial-and-error process to balance immediate profits with long-term stability.

4.2. Comparative methods and evaluation metrics

In our research, we examined various algorithmic strategies from prior investigations, incorporating a range of rule-based techniques. Additionally, we incorporated an advanced method derived from contemporary research in RL to serve as a benchmark for cutting-edge performance. The methods selected for comparison are outlined as follows:

- **Random**: This strategy involves choosing N_{env} diverse environments from historical records at random each day to update the RL agent.
- **Cluster-Based Update**: This strategy entails conducting daily evaluations of the previous day's market sentiment to identify its cluster. Subsequently, the agent updates its status by utilizing data from that specific cluster.
- **Trend-Similar Update**: This method analyzes the market sentiment from the previous day and selects data with similar market trends within the last 90 days for updating.
- **FinRL** [37]: This trading system is open-source and incorporates several traditional RL algorithms. Our module has been developed using this framework, making it essential to conduct a comparative analysis between this established system and our own methodology.
- **MVMM** [38]: The Multi-Agent Virtual Market Model (MVMM) combines multiple Generative Adversarial Networks (GANs) to generate varied market trends. This setup allows agents to learn and adapt by engaging with complex virtual market dynamics created by the MVMM.
- **CRL-SSMCN** [2]: The method introduces a Contextual RL framework that uses financial market sentiment indicators to customize trading strategies based on market conditions.

In this experiment, we adopted a comprehensive set of performance evaluation metrics, encompassing both profitability indicators and risk measures. These metrics collectively assess the strategy's earning potential and its risk-adjusted stability, detailed as follows.

- **Profit and loss (PnL)**: This is the daily net profit or loss measured in TAIEX points, indicating the strategy's overall profitability.
- **Mean Average Position (MAP)**: This represents the daily average inventory held, reflecting the strategy's exposure to inventory risk.
- **PnL-MAP Ratio**: By dividing PnL by MAP, this metric offers insight into the efficiency of profit generation relative to the risk taken.
- **Sharpe Ratio (SR)**: This ratio of profit to the standard deviation evaluates the profit's consistency.
- **Maximum Drawdown (MDD)**: Identifies the greatest loss from a peak to a trough in the strategy's daily performance.
- **Final Inventory (INV)**: The mean inventory level at the end of the trading day.

4.3. System parameter tuning and evaluation

In this experiment, our primary goal is to determine the optimal parameter K, which is used to decide when sudden concept drift occurs. We also aim to identify the most suitable number of days D, to ascertain whether sudden concept drift has happened. The purpose of this is to ensure that our agent can maintain real-time and effective trading strategies while engaging in market-making activities. Furthermore, by determining these parameters, we expect the agent to not only enhance its profitability but also better mitigate risks when facing market changes. We observed the impact and correlation of different settings for the sudden concept drift threshold hyperparameter K and the update frequency hyperparameter D on training performance. We conducted eighteen different experiments using six distinct values of K and three different values of D. The range for K varied from 0.025 to 0.05, while the values for D were set at 1, 3, and 5. The experimental results are presented in Table 2.

Based on the results presented in Table 2, we can observe the impacts of various combinations of the sudden concept drift threshold (K) and the update frequency (D) on the training performance of the agent. From the experimental outcomes, it is evident that the setting with a sudden concept drift threshold of $K = 0.035$ and an update frequency of $D = 3$ days shows remarkable performance across several metrics. Specifically, this setting results in the highest values for the PnL-MAP, Sharpe Ratio, and has the lowest MAP and MDD, indicating a robust balance between profitability and risk management. The optimal performance of this configuration suggests that it effectively captures the market dynamics, adjusting the trading strategy in a timely manner in response to sudden concept drift, thereby maintaining the agent's competitiveness in the market-making activities.

The performance metrics for the $K = 0.035$ and $D = 3$ settings confirm their superiority in optimizing the agent's market-making

Table 2
Impact of sudden concept drift threshold and update frequency on training performance.

Period	K	D	Evaluation metrics					
			PnL	MAP	PnL-MAP	MDD	SR	INV
201 901 ~ 201 906	0.025	1	19.223	16.370	3.499	1042.681	0.088	53.001
	0.030	1	41.210	16.001	9.039	518.616	0.178	29.203
	0.035	1	84.184	13.054	9.201	478.069	0.309	18.499
	0.040	1	−5.388	12.608	5.456	498.608	0.188	18.244
	0.045	1	−75.072	37.058	5.194	1443.379	0.116	57.639
	0.050	1	**104.403**	18.827	9.565	686.367	0.262	33.119
	0.025	3	23.733	14.522	3.735	564.566	0.122	42.129
	0.030	3	47.754	18.771	8.383	475.496	0.209	25.680
	0.035	3	64.992	**5.090**	**19.019**	**195.626**	**0.360**	**8.172**
	0.040	3	11.495	6.198	17.592	260.385	0.195	8.572
	0.045	3	−50.499	9.092	5.343	397.987	0.128	14.089
	0.050	3	−6.197	11.353	4.567	461.260	0.030	17.525
	0.025	5	26.527	16.362	6.725	1031.081	0.092	51.341
	0.030	5	30.532	16.948	8.812	554.228	0.168	24.132
	0.035	5	69.516	12.314	8.394	437.037	0.310	17.985
	0.040	5	−40.705	10.291	8.246	453.056	0.159	14.905
	0.045	5	−26.524	10.989	10.801	466.751	0.266	15.194
	0.050	5	−66.812	17.844	3.479	697.036	0.111	29.617

strategies. Although the PnL figures for this setting are not the absolute highest, they are significantly positive, indicating the agent's effectiveness in generating profits across diverse market conditions. This performance is underpinned by an exceptionally low MAP of 5.090, which evidences the agent's ability to maintain a balanced market position, thereby mitigating the risk of significant losses through overexposure. Furthermore, the lowest MDD among the configurations signifies superior risk management, crucial for the agent's long-term operational sustainability. The standout Sharpe Ratio in this setting underscores not merely the high returns but also their consistency, reflecting the agent's adeptness at real-time strategy adjustments. This holistic performance validation accentuates the $K = 0.035$ and $D = 3$ parameters as the optimal choice for ensuring the agent's robust adaptability and sustained profitability in the face of market dynamics.

In conclusion, the experiment clearly demonstrates that the parameter setting of $K = 0.035$ and $D = 3$ days stands out as the most effective configuration for the Agent in terms of maintaining a balanced trade-off between profitability and risk management. This finding is crucial for enhancing the agent's ability to adapt to dynamic market conditions, thereby ensuring more stable and consistent performance in its market-making activities. Consequently, we recommend adopting this parameter setting for the agent to optimize its trading strategies in the face of sudden concept drift.

4.4. System performance evaluation

In this experiment, we attempt to compare our method with other existing methods, including three rule-based methods and three learning-based methods. The experimental results are presented in Tables 3 and 4. In these tables, we show the testing results for the call option conditions and calculate the average daily test results over periods of six months and two months, respectively.

In the experimental results presented in Table 3, it is evident that our method excels across several key metrics. Our approach achieved the highest PnL at 122.499 among all the methods, with a high PnL indicating our strategy's capability to discern more profit opportunities in the market. In comparison, other methods such as Cluster-Based and CRL-SSMCN reported PnLs of 99.925 and 84.333, respectively, demonstrating our method's superior ability to capture profit opportunities. Turning to MAP, our method registered a performance of 1.702, which is on the lower side, signifying that our approach does not accumulate excessive inventory during market-making. Although the MAP value of CRL-SSMCN is close to ours, our method exhibits better

Table 3
Performance of comparison methods.

Period	Method	Evaluation metrics					
		PnL	MAP	PnL-MAP	MDD	SR	INV
201 901 ~ 201 906	Random	61.714	4.277	14.553	129.749	0.289	3.760
	Cluster-Based	99.925	6.833	14.927	253.430	0.294	8.042
	Trend-Similar	64.157	5.888	14.029	217.475	0.303	7.506
	FinRL	38.118	10.322	3.139	420.925	0.144	9.096
	MVMM	3.274	7.635	1.097	350.664	0.294	7.349
	CRL-SSMCN	84.333	1.776	47.455	**73.461**	1.071	**1.688**
	CL&KD	**122.499**	**1.702**	**73.297**	110.969	**1.136**	2.013

overall performance when combining other metrics like PnL. Regarding PnL-MAP, our method scored 73.297, significantly surpassing all other approaches, indicating a well-balanced trade-off between profitability and inventory risk, maintaining stable performance under fluctuating market conditions. As for MDD, our method recorded 110.969, which, despite not being the lowest among all methods, is considered acceptable given the high PnL. CRL-SSMCN outperformed in this metric at 73.461, but its PnL was much lower than ours, suggesting good risk control but weaker profitability. Concerning the Sharpe ratio, our method topped with 1.136, indicating the best risk-adjusted return rate among all strategies, hence showcasing superior risk-adjusted returns compared to all other methods. Finally, for Inventory, our method's close inventory performance was not the best but showed relatively low values, indicating our strategy's effectiveness in gradually clearing out positions towards market close to minimize overnight risks.

These results demonstrate our method's effective adaptability to market changes, achieving more accurate predictions and higher profitability. Our method incorporates a sudden drift detector, suggesting our model's ability to identify market changes in market sentiment. For minor market changes, we refine the model through knowledge distillation, ensuring it learns from existing knowledge and adapts to gradual concept drift. For significant market changes, we employ curriculum learning to retrain the model, enabling quicker adaptation to new market conditions rather than merely relying on past strategies. This dynamic adaptation method is a notable advancement over traditional methods. Knowledge distillation enhances model performance, improving its efficacy in similar market conditions, while curriculum learning allows for retraining through a gradual complexity increase, preventing overfitting through excessive training. Compared to methods like Random, Cluster-Based, and Trend-Similar, our approach demonstrates advanced market adaptability adjustments and updated strategies, capturing and responding to market changes more accurately. Against traditional algorithm-based FinRL, virtual market-dependent MVMM, and market sentiment-integrating CRL-SSMCN, our method shows greater flexibility and adaptability, thereby outperforming multiple key metrics.

In Table 4, we delve further into the performance of each method during different market trend periods: January 2019 to February 2019 as the consolidation period, March 2019 to April 2019 as the uptrend period, and May 2019 to June 2019 as the downtrend period. During the consolidation period, the market did not exhibit a clear upward or downward trend, and trading was relatively stable. Our method achieved the highest PnL during this period, significantly surpassing other methods, which indicates our method's capability to capture effective trading opportunities in a relatively stable market. Additionally, our method's Sharpe ratio was the highest, demonstrating that during the consolidation period, our approach not only excelled in profitability but also achieved the highest return per unit of risk. During the uptrend period, our method demonstrated good adaptability and profitability. Although slightly lower in PnL compared to Cluster-Based, our method showed a more balanced and stable performance considering risk factors. It also reached the highest PnL-MAP during this period, proving our method's efficiency in leveraging profit opportunities while maintaining lower position holdings for higher gains. During

Table 4
Segmented performance analysis of comparison methods.

Period	Method	Evaluation metrics					
		PnL	MAP	PnL-MAP	MDD	SR	INV
201 901 ~ 201 902	Random	83.339	4.061	20.519	148.173	0.578	3.688
	Cluster-Based	123.979	7.190	17.243	256.680	0.246	9.375
	Trend-Similar	36.128	6.789	14.087	299.083	0.354	9.531
	FinRL	157.330	11.720	13.424	429.570	0.250	9.910
	MVMM	−9.088	7.838	−1.159	414.184	0.349	7.281
	CRL-SSMCN	150.919	2.498	60.411	**106.959**	1.211	**2.094**
	CL&KD	**186.177**	**2.237**	**83.226**	120.853	**1.354**	2.294
201 903 ~ 201 904	Random	52.727	4.367	12.074	122.092	0.168	3.789
	Cluster-Based	**110.309**	5.928	18.608	196.441	0.428	6.895
	Trend-Similar	87.122	5.061	17.216	154.132	0.360	5.947
	FinRL	−7.210	8.440	−0.854	350.830	0.130	7.800
	MVMM	34.266	5.738	5.971	213.681	0.363	4.711
	CRL-SSMCN	65.697	**1.129**	58.170	**29.266**	**1.075**	1.500
	CL&KD	102.694	1.154	**88.990**	110.468	1.069	**1.456**
201 905 ~ 201 906	Random	52.727	4.367	12.074	122.092	0.168	3.789
	Cluster-Based	70.071	7.423	9.440	306.292	0.202	8.065
	Trend-Similar	64.779	5.956	10.877	212.234	0.207	7.364
	FinRL	−15.530	11.010	−1.411	482.130	0.070	9.690
	MVMM	−16.779	9.318	−1.801	432.016	0.181	9.974
	CRL-SSMCN	47.856	1.814	26.383	**89.038**	0.953	**1.538**
	CL&KD	**89.547**	**1.796**	**49.859**	103.347	**1.023**	2.324

the downtrend period, while many trading strategies struggled, our method still achieved a positive PnL, the highest among all methods. Additionally, our method's Sharpe Ratio remained high during the downtrend, indicating that even in a falling market, our approach could adjust strategies effectively to manage risk and secure relatively stable returns.

Our method integrates a sudden drift detector and timely model adjustments to promptly adapt to changes in market environments, whether in consolidation, uptrend, or downtrend periods. Knowledge distillation helps refine and optimize the model under various market conditions, while curriculum learning efficiently retrains the trading model when significant market changes are detected.

4.5. Ablation study and key design analysis

In this experiment, we aim to explore the effects of market sentiment when trained with various sentiment features. We use a variety of settings to train our TSSMSN, integrating market sentiment data, denoted as X^{MS}, into the network to generate a range of sentiment embeddings. These embeddings are then utilized to train a downstream trading agent, and the performance is evaluated through backtesting. We categorize the sentiment features into seven distinct groups and establish a baseline by comparing their results with an agent trained without market sentiment information. The variables are defined as follows: "B" represents PCR_B, "S" represents PCR_S, and "O" represents OI. By comparing the results across these seven experimental groups, we can derive more granular insights into how each aspect of market data contributes to market sentiment. The experimental results are presented in Tables 5 and 6. In these tables, we show the ablation study results for the call option conditions and calculate the average daily test results over periods of six months and two months, respectively. Each column corresponds to a distinct evaluation metric, whereas each row delineates the auxiliary tasks employed across various groups.

Beginning with Table 5, we can delve deeper into the impact of various sentiment feature combinations on the performance of trading strategies. Looking at PnL, the BS and BSO combinations show better profitability, indicating that integrating market sentiment information helps the model capture market dynamics, thereby increasing profits. In terms of MAP, the BSO strategy performs the best, suggesting that the BSO combination can better adjust positions to balance risk and reward. The PnL-MAP ratio further highlights the efficiency of the BSO strategy

Table 5
Performance of different combinations of auxiliary tasks in the TSSMSN.

Period	Method	Evaluation metrics					
		PnL	MAP	PnL-MAP	MDD	SR	INV
201 901 ~ 201 906	None	−16.689	21.099	−1.138	80.196	0.056	27.463
	B	11.061	6.606	1.692	39.755	0.012	8.417
	S	−25.622	9.840	−2.947	66.461	−0.093	12.872
	O	33.164	3.817	8.698	35.844	−0.001	3.683
	BS	**75.246**	4.358	16.998	42.964	0.002	4.307
	BO	53.942	4.122	12.362	45.121	0.076	4.294
	SO	58.196	4.866	11.448	41.714	**0.090**	6.229
	BSO	59.875	**2.010**	**29.635**	**31.873**	0.089	**1.734**

Table 6
Segmented performance analysis of auxiliary task combinations in the TSSMSN.

Period	Method	Evaluation metrics					
		PnL	MAP	PnL-MAP	MDD	SR	INV
201 901 ~ 201 902	None	29.927	29.647	1.009	116.274	0.036	37.469
	B	43.009	6.986	6.156	**27.642**	0.022	7.891
	S	26.328	11.182	2.354	41.078	−0.030	14.203
	O	55.384	3.807	14.549	27.923	−0.023	3.813
	BS	**175.155**	4.485	39.050	36.311	−0.023	4.297
	BO	105.695	4.860	21.748	37.641	**0.073**	5.219
	SO	120.527	5.529	21.797	33.348	0.065	7.047
	BSO	110.573	**2.289**	**48.312**	29.437	0.028	**1.531**
201 903 ~ 201 904	None	−47.655	18.330	−2.600	49.122	0.055	24.658
	B	4.668	5.301	0.881	38.942	0.003	7.197
	S	−48.688	8.371	−5.816	68.246	−0.127	11.658
	O	33.938	3.816	8.893	38.183	0.029	3.421
	BS	34.201	4.281	7.990	42.678	−0.010	4.237
	BO	27.796	3.633	7.651	47.836	**0.098**	3.566
	SO	28.711	3.705	7.750	43.897	0.093	4.395
	BSO	**55.263**	**1.659**	**33.312**	**34.401**	0.075	**1.434**
201 905 ~ 201 906	None	−24.766	16.782	−1.476	80.871	0.074	21.987
	B	−8.923	7.567	−1.179	50.487	0.013	10.038
	S	−45.772	10.169	−4.501	85.547	−0.113	12.962
	O	14.179	3.826	3.706	40.064	−0.012	3.833
	BS	33.263	4.330	7.681	48.702	0.036	4.385
	BO	**36.952**	3.994	9.252	48.613	0.059	4.244
	SO	35.782	5.454	6.561	46.452	0.108	7.346
	BSO	22.769	**2.122**	**10.729**	**31.409**	**0.152**	**2.192**

in achieving high returns with smaller position changes. This increase in efficiency indicates that the BSO strategy can effectively enhance profits while controlling risks. The analysis of the Sharpe Ratio supports this point, with the higher values for BSO and BO indicating that these strategies achieve higher excess returns while adjusting for risk.

The low value in MDD for the BSO combination indicates its strong risk control ability, meaning that even in the face of rapid market fluctuations, the BSO strategy can quickly adjust and minimize losses. The Inv shows that the BSO strategy effectively reduces open positions at the end of trading, helping to lower overnight risk.

Additionally, observing Table 6 allows us to examine different market segments. By evaluating the performance of all strategies across these market trends, we can discern their adaptability under varying conditions. During the consolidation period, the BSO strategy excels across several metrics, particularly in PNL-MAP and MAP, indicating its ability to maintain strong performance amidst high market uncertainty. Moving to the rising market period despite the overall positive market performance, the BSO strategy stands out again in PnL and PnL-MAP metrics, showcasing its advantage in leveraging comprehensive market sentiment data. Regarding the declining period, although BSO is not the best in PnL, it still shows better stability in PnL-MAP and MDD, reflecting its resilience under market downturn pressures. In summary, the BSO strategy demonstrates adaptability and robustness across different market trends, proving the effectiveness of integrating multidimensional market sentiment data into the strategy. Regardless of whether the market is consolidating, rising, or falling, the BSO

Table 7
Performance comparison of methods on put options.

Period	Method	Evaluation metrics					
		PnL	MAP	PnL-MAP	MDD	SR	INV
	Random	27.662	9.207	2.989	457.146	0.122	16.699
	Cluster-Based	49.863	7.450	8.407	328.030	0.179	12.964
201 901	Trend-Similar	106.465	5.886	15.607	256.356	0.276	10.906
~	FinRL	48.334	7.753	6.617	351.298	0.176	11.998
201 906	MVMM	126.813	5.733	23.322	219.123	0.333	7.773
	CRL-SSMCN	146.995	3.580	40.585	**156.588**	0.671	**4.840**
	CL&KD	**159.062**	**3.233**	**44.445**	180.869	**0.716**	5.964

Table 8
Performance comparison of methods on TEO options.

Period	Method	Evaluation metrics					
		PnL	MAP	PnL-MAP	MDD	SR	INV
	Random	26.272	5.003	13.831	210.775	0.101	6.457
	Cluster-Based	70.414	3.368	29.860	124.710	0.360	4.308
201 901	Trend-Similar	70.966	4.194	12.504	142.995	0.102	5.909
~	FinRL	45.703	**2.793**	16.256	96.314	0.144	**2.801**
201 906	MVMM	45.088	20.468	10.356	736.825	0.237	33.889
	CRL-SSMCN	72.160	3.471	20.721	118.506	0.326	5.767
	CL&KD	**88.456**	3.094	**29.970**	**94.566**	**0.369**	6.112

strategy consistently performs well, illustrating its reliability in various market conditions.

In conclusion, the BSO method, by comprehensively considering multiple dimensions of market sentiment indicators, provides more holistic and precise trading signals. This not only enhances the model's profit-making capability but also strengthens risk control, thereby proving to be particularly effective in improving the performance of market-making strategies.

4.6. Experiments on different commodities

In this experiment, we intend to apply our method to Put options to see if its performance also surpasses other methods. The selection of training and testing periods is the same as that chosen for the call options. Detailed information about the dataset is shown in 4.1. The experimental results are presented in Table 7. In this table, we show the testing results for the put option conditions and calculate the average daily test results over periods of six months.

Based on experimental results, our method demonstrates significant advantages across nearly all key metrics. Particularly in metrics such as PnL, PnL-MAP, and Sharpe Ratio, our method not only surpasses all baselines but also maintains a level comparable to the best comparative method in maximum drawdown, highlighting the stability and reliability of our approach.

Further analysis reveals that while methods such as Trend-Similar Update and Cluster-Based Update offer non-specific learning strategies, they are noticeably inadequate when facing significant market changes. In contrast, our method can promptly adjust learning strategies by detecting sudden concept drift, whether through fine-tuning via knowledge distillation or through comprehensive retraining via curriculum learning upon detecting significant changes, significantly enhancing the model's adaptability to market dynamics.

When compared with other advanced methods like MVMM and CRL-SSMCN, our approach still exhibits significant competitive strength. Although the MVMM method utilizes GANs to generate diverse market trends, it falls short in quickly adjusting its strategies to respond to sudden market changes. Meanwhile, while CRL-SSMCN has made significant optimizations in model market sensitivity and understanding, it remains slightly inferior to our method in rapidly responding to concept drift. This further proves the importance of combining sudden drift detector with flexible learning strategies, especially in highly volatile markets. In conclusion, these experimental results not only confirm the effectiveness of our method in handling put options but also showcase its superiority over other advanced methods, particularly in terms of rapid adaptation and precise identification of market changes.

4.7. Profitability evaluation for additional dataset

In this experiment, we aim to apply our method to various datasets to evaluate its performance against other methods. We utilized tick data from TEO, listed on the TAIFEX. TEO is the second most actively traded product in the Taiwan options market. The training and testing periods were selected to match those used for the TXO. The experimental results are presented in Table 8, which displays the testing outcomes for the TEO and the average daily test results over six-month periods.

Based on experimental results, our method demonstrates significant advantages across nearly all key metrics. Particularly in metrics such as PnL, PnL/MAP, and Sharpe Ratio, our method not only surpasses all baselines but also maintains a level comparable to the best comparative method in MDD, highlighting the stability and reliability of our approach. These results demonstrate significant advantages in both stability and risk management. This further proves the importance of combining sudden drift detector with flexible learning strategies, especially in highly volatile markets. The ability to swiftly detect and adapt to sudden market changes is crucial in options trading, where unexpected fluctuations can greatly impact performance. By integrating a sudden drift detector, our method ensures that the model remains effective, adjusting its parameters dynamically to reflect the latest market conditions. Our method's capacity to quickly identify and adapt to these market changes means that it can maintain high performance even as market dynamics evolve. Regarding the final inventory, our method shows a relatively high value. This might indicate a propensity for holding positions longer than necessary or difficulties in liquidating positions efficiently. The adaptive nature of our method, which excels in dynamically adjusting to market changes, might sometimes result in overfitting to recent market conditions, causing an accumulation of positions. This could be addressed by incorporating more stringent inventory control mechanisms or enhancing the liquidation strategy to ensure positions are closed more efficiently.

In conclusion, the experimental results demonstrate the effectiveness of our method in navigating the complexities of the TEO market. This further substantiates the robustness of our method across various datasets, showcasing its broad applicability. The consistency of the results obtained from diverse datasets underscores the reliability and efficiency of our approach in addressing challenges beyond the TXO market.

5. Conclusion

In this study, we present a novel reinforcement learning framework with environmental sentiment awareness for market-making that combines knowledge distillation and curriculum learning to tackle the challenge of identifying and resolving the traditional concept drift problem in financial markets. Our approach enables trading models to dynamically adapt to both daily gradual concept drift and sudden concept drift, restructuring themselves during significant market changes.

The knowledge distillation technique is adopted to intelligently guide the selection of historical environments for fine-tuning trading models, thereby addressing daily gradual concept drift. This adaptability is crucial for maintaining performance over time, ensuring effective trading and risk management. For sudden concept drift, the sudden drift detector and curriculum learning framework are further utilized to enhance learning efficiency across diverse time segments and extensive learning environments. Through the sudden drift detector, significant market changes could be identified promptly. By gradually increasing the training task complexity, curriculum learning helps reset the agent's learning process efficiently, enhancing adaptability to significant market changes. This dual approach effectively handles concept drift, boosting the resilience and performance of the trading strategy. The experimental results with TAIEX Options data demonstrate that our method achieves better profitability, increasing PnL by 38.17%, while maintaining the same level of inventory risk. This also validates that our proposed market-making strategy, based on a reinforcement learning framework with environmental sentiment awareness, and incorporating knowledge distillation and curriculum learning, effectively enhances trading performance by identifying and resolving the problem of concept drift.

In conclusion, integrating market sentiment analysis with reinforcement learning provides a powerful enhancement to high-frequency market-making strategies. Our study highlights the importance of managing concept drift to ensure that the trading model can adapt and restructure itself amid market changes. Beyond this, several promising directions remain for future exploration. From an application perspective, different trading scenarios, asset types, and funding costs could be examined to assess how these factors influence model behavior and profitability. From a methodological standpoint, aspects such as the choice of discount factor and reward design can be further refined to achieve more precise control over agent behavior and performance. Moreover, future research could incorporate multi-agent systems, where trading agents interact and compete, offering deeper insights into market dynamics. Such an approach would better simulate real-world trading environments and enhance agent performance through collective learning and competition.

CRediT authorship contribution statement

Chang-An Wang: Writing – original draft, Software, Methodology, Formal analysis, Data curation. **Szu-Hao Huang:** Supervision, Resources, Funding acquisition. **Chiao-Ting Chen:** Writing – review & editing, Project administration. **Yi-Tang Fang:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The data that has been used is confidential.

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